7.7 Activities: metric system and area

← Unit 7.7 · Practice · Games

This unit is best done with real measuring tools in his hands. Paper practice comes after.

1. Measure the house

You need: a tape measure, a ruler, a notebook.

How: Make a list together: the length of his bed, the height of the door, the width of a book, the thickness of a coin, the length of the hallway. Before measuring, he guesses (and writes the guess down). Then he measures and writes each result in two units: 1,92 m = 192 cm; 2 mm = 0,2 cm.

Talk about it: "Which unit makes sense for each thing? Would you measure the hallway in mm?" "Were your guesses closer by the end?"

2. Kitchen conversions

You need: a measuring jug, kitchen scales, packaging.

How: Find packs with grams, kilos, litres and ml. Write each one the other way: 1,5 kg = 1 500 g; 330 ml = 0,33 l. Then weigh or measure: pour 250 ml of water. Weigh *een ons* (100 g) of pasta. How many glasses of 0,2 l in a 1,5 l bottle?

Talk about it: "Why does *een ons* still exist when nobody writes it on packets?"

3. Fence the garden

You need: a garden, balcony or park; a long string or tape measure.

How: Pretend you're putting a fence round a flower bed, the garden, or the sandpit. Measure the sides and add them up for the perimeter. If fencing costs € 15 per metre, what does it cost? Draw the shape on squared paper too, with 1 square = 1 m.

Talk about it: "If we made the garden twice as long, would the fence be twice as long?"

4. Build a square metre

You need: newspapers, tape, a ruler, scissors.

How: Tape newspaper sheets together and cut a square of exactly 1 m by 1 m. That's 1 m². How many children can stand on it? Then cut a 10 cm × 10 cm square (100 cm²) and see how many fit along one side (10) and how many on the whole square metre (10 × 10 = 100). So 1 m² = 100 × 100 cm² = 10 000 cm².

Talk about it: "Did you expect 1 m² to be 100 cm²? Why isn't it?" "About how many m² is your bedroom floor?"

5. Floor plan of my room (*plattegrond*)

You need: squared paper, a tape measure.

How: Measure his room (to the nearest 10 cm) and draw it on squared paper with 1 square = 50 cm or 1 m. Add the bed and desk as rectangles. Work out the area of the floor, and the area that is still free. If the room isn't a rectangle, split it into rectangles.

Talk about it: "If we wanted new carpet, how much would we need to buy? Why might we buy a bit more?"

6. Same area, different shape

You need: 12 small square tiles, sugar cubes or Lego plates, or squared paper.

How: Make as many different rectangles as possible with all 12 squares (1 × 12, 2 × 6, 3 × 4). They all have area 12. Count the perimeter of each. Which has the biggest perimeter? The smallest?

Talk about it: “Same area, different perimeter. How is that possible?”

Conversation prompts for any day

- “How long is a kilometre? Which places are about 1 km from our house?”
- “Is a metre longer or shorter than your arm span? Let’s check.”
- “When would you need omtrek and when oppervlakte?” (Fence vs grass seed, frame vs paint.)